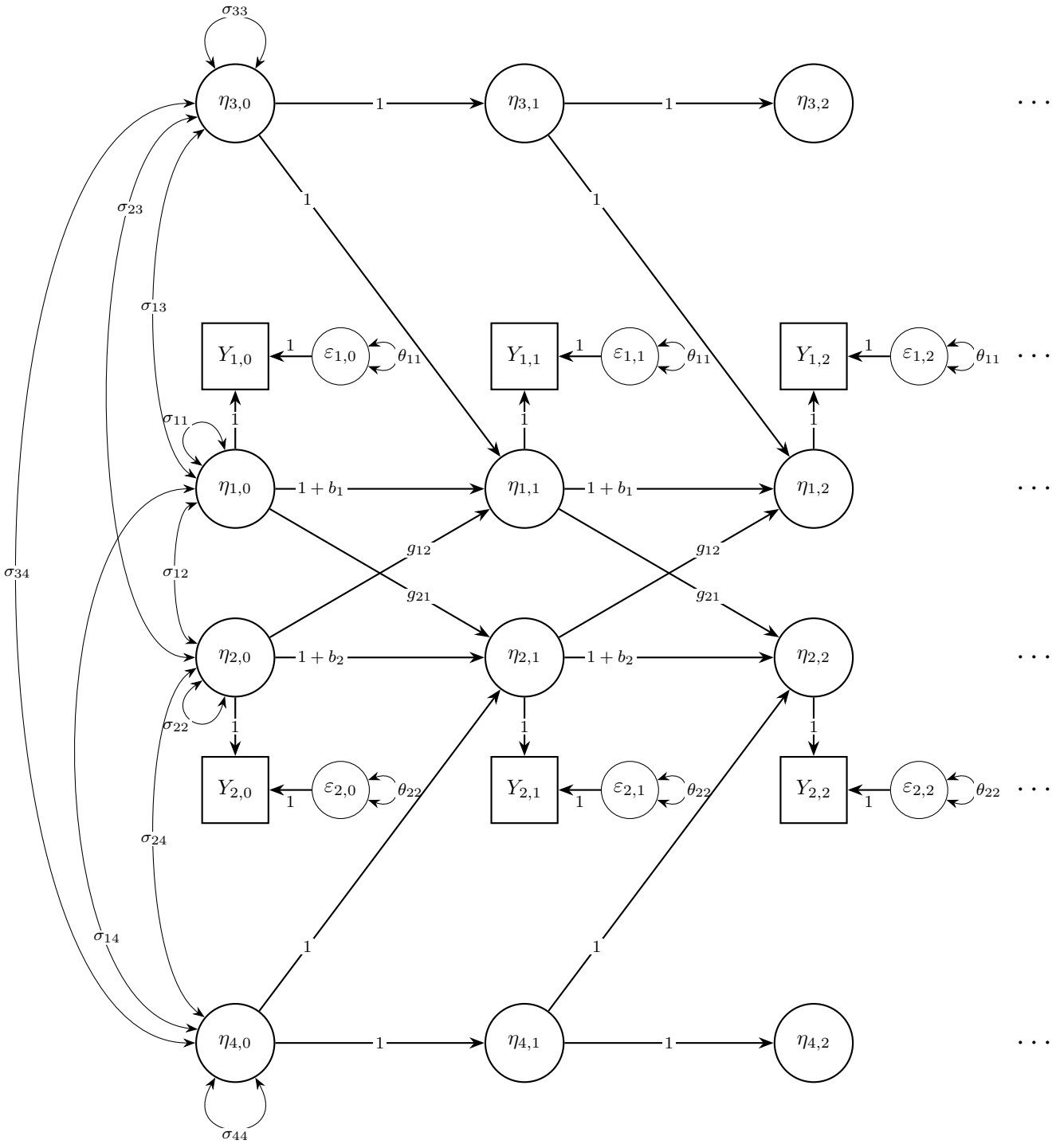


$\boldsymbol{\eta}_t = (\eta_{1,t}, \eta_{2,t}, \eta_{3,t}, \eta_{4,t})'$, $\quad \mathbf{y}_t = (Y_{1,t}, Y_{2,t})'$; no process noise

true levels $\eta_{1,t}$ and $\eta_{2,t}$; constant change components $\eta_{3,t} = \eta_3$ and $\eta_{4,t} = \eta_4$ for all t

$$\Delta\eta_{1,t} = \eta_3 + b_1\eta_{1,t-1} + g_{12}\eta_{2,t-1}, \quad \Delta\eta_{2,t} = \eta_4 + b_2\eta_{2,t-1} + g_{21}\eta_{1,t-1}$$

$$\boldsymbol{\beta} = \begin{pmatrix} 1+b_1 & g_{12} & 1 & 0 \\ g_{21} & 1+b_2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \qquad \boldsymbol{\Theta} = \begin{pmatrix} \theta_{11} & 0 \\ 0 & \theta_{22} \end{pmatrix}$$

$$\boldsymbol{\eta}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\mu}_0 = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \end{pmatrix}, \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} & \sigma_{14} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} & \sigma_{24} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} & \sigma_{34} \\ \sigma_{41} & \sigma_{42} & \sigma_{43} & \sigma_{44} \end{pmatrix}$$